# CSE 115L - Lab 07 Task Sheet: Completed Answers

North South University | Nested if and Multiple-Alternative Decisions | Fall 2026

**Note:** N is digits 4 to 7 of your own student ID. Fill in your N-based answers yourself, type and run every program with -Wall on, and show the instructor all outputs in the room. Name, ID, date and signatures are yours to add.

## Guided Tasks G7.1 and G7.2 (no marks, but required)

**G7.1:** blank (1) = **90**, (2) = **3.3**, (3) = **80**, (4) = **0.0**.

A mark of 75 gets **0.0**. It fails every test (75 < 80), so it reaches the final else, but 75 should be C = 2.0. The ladder skips the 77, 73, 70, 67 and 60 bands.

**G7.2:** corrected structure

```
if (registered) {
    if (mark >= 60)
        printf("Result: PASS\n");
} else
    printf("Not registered for this course.\n");
```

With registered = 1 and mark = 45 nothing prints. That is wrong because a registered student who failed should be told so. The inner if needs its own `else printf("Result: FAIL\n");`.

## E7.1 - Which branch runs? (Easy, 1 mark)

| Mark | Branch number | Grade printed |
|---|---|---|
| 93 | 1 | A |
| 92 | 2 | A- |
| 60 | 10 | D |
| 59 | 11 | F |
| N % 101 | (compute, see key) | (see key) |

Key for N % 101: 93+ = branch 1 (A); 90-92 = 2 (A-); 87-89 = 3 (B+); 83-86 = 4 (B); 80-82 = 5 (B-); 77-79 = 6 (C+); 73-76 = 7 (C); 70-72 = 8 (C-); 67-69 = 9 (D+); 60-66 = 10 (D); below 60 = 11 (F). Example: N = 1234 gives 1234 % 101 = 22, so branch 11, grade F.

## E7.2 - Negative, zero or positive (Easy, 1 mark)

```
#include <stdio.h>
int main(void)
{
    double x;
    printf("Enter a number: ");
    scanf("%lf", &x);
    if (x < 0)       printf("Negative\n");
    else if (x == 0) printf("Zero\n");
    else             printf("Positive\n");
    return 0;
}
```

## M7.1 - The full grade report (Mediocre, 1 mark)

Type Worked Example 7.1 and run it on 95, 91, 88, 85, 81, 78, 75, 71, 68, 63 and 42. After the printf that prints the grade, add a one-alternative if (not a twelfth branch):

```
if (mark >= 90)
    printf("Distinction\n");
```

## M7.2 - Validate before you classify (Mediocre, 1 mark)

Declarations and scanf stay as in 7.1. The ladder and the final printf go inside the outer if:

```
if (mark >= 0 && mark <= 100) {
```

```
    /* the 11-branch ladder from Worked Example 7.1 goes here */
    printf("Mark %3d Grade %c%c Grade point %.1f\n",
           mark, letter, modifier, point);
} else
    printf("Invalid mark: %d (must be 0-100)\n", mark);
```

Test with -5 and 101 (both print the error) and with N % 101 (prints a grade). Keep the final printf inside the if, otherwise bad data would print uninitialised variables.

## M7.3 - Two variables, four outcomes (Mediocre, 2 marks)

```
#include <stdio.h>
int main(void)
{
    int mark, att;
    printf("Enter mark and attendance %%: ");
    scanf("%d %d", &mark, &att);
    if (mark >= 60) {
        if (att >= 70) printf("Passed\n");
        else           printf("Failed on attendance\n");
    } else {
        if (att >= 70) printf("Failed on mark\n");
        else           printf("Failed on both\n");
    }
    return 0;
}
```

| Input (mark att) | Output |
|---|---|
| 75 80 | Passed |
| 75 50 | Failed on attendance |
| 50 80 | Failed on mark |
| 50 50 | Failed on both |

Show the instructor all four outcomes actually running (Rule 5).

## C7.1 - The ladder with a hole (Challenging, 2 marks)

**Marks that fall through:** 0 to 64 inclusive (65-69 give D).

**What prints:** "Grade " followed by whatever byte letter happened to hold. This is undefined behaviour, so it may be junk or a blank.

**Does -Wall warn?** Maybe. -Wmaybe-uninitialized is part of -Wall, but GCC usually detects it only with optimisation on (-O1 or higher). At the default -O0 in Code::Blocks it is often silent, so do not rely on it. Check on your machine and report what you see.

**Fix 1 (add a branch):** add `else letter = 'F';` after the last else if.

**Fix 2 (initialise):** `char letter = 'F';` before the ladder.

**Keep:** the else. It states explicitly what happens for every mark, while an initial value can silently mask a missing case later. Show both fixes building (Rule 5).

## C7.2 - Make the ladder lie (Challenging, 2 marks)

**The one character:** in the mark >= 83 line, delete the = to give `mark > 83`.

**Why it passes silently:** it builds with no warning and every mark still gets a grade.

**Mark that exposes it: 83**. It now falls to the B- branch and prints B- 2.7, but the correct answer is B 3.0.

**How to catch it:** boundary testing. For every band test the lower bound and one below it (93/92, 90/89, 87/86, 83/82, 80/79, 77/76, 73/72, 70/69, 67/66, 60/59) and compare each result with the table in section 1.3. Mid-band marks such as 85 still pass, so they will not reveal the fault.

**Before you leave:** fill in name, ID, section and date; get each task checked and the sheet signed in the room. Nothing is marked afterwards.